# ISE501-INTRODUCTION TO OPERATIONS RESEARCH

## Multi-Plant Production and Supply Chain Optimization Under Tariff and Environmental Policies

## Abstract

This project develops a linear programming model for **NovaRide**'s global electric scooter production network. The system features two manufacturing plants (China and India), two U.S. markets (East and West), four essential components (battery, motor, PCB, chassis), and two component suppliers. The model minimizes total cost encompassing production, shipping, component procurement, tariffs, inventory, shortages, and emissions, while meeting monthly demand and respecting operational constraints. Numerical experiments include India-only and China-only production, tariff increases, supplier disruptions, production policy comparisons, and service-level requirements. The Base Case reveals India's cost-efficiency and China's role as a flexible WEST-serving plant. Scenario analysis demonstrates the tradeoffs between cost, resilience, service levels, and sustainability.

## Introduction

Global supply chains increasingly rely on distributed production systems that must respond to fluctuating demand, tariff structures, supplier constraints, and sustainability targets. For companies operating internationally, optimizing such systems is critical to achieving cost efficiency and operational reliability. This project focuses on NovaRide's scooter manufacturing network, which operates two plants (China and India), sources components from two suppliers, and distributes finished goods to East and West U.S. markets. The problem is modeled as a linear program solved using GAMS. Key objectives include:

- ➢ Minimizing total operational cost
- ➢ Meeting monthly market demand
- ➢ Respecting plant capacity and supplier availability
- ➢ Evaluating sensitivity to tariffs, disruptions, and service levels

## Problem Description

• Plants in India and China have different production costs, capacities, and emission rates.
• Suppliers have limited monthly availability for each component.
• East and West U.S. markets show different seasonal demand.
• Tariffs apply to imported components and finished goods.
• The company aims to limit total $CO_2$ emissions while maintaining a desired service level.

The model optimizes how much to produce, ship, and procure monthly while meeting demand and sustainability requirements.

## Model Formulation

### a) Decision Variables

| Variable | Description | Units |
|---|---|---|
| q[p,t] | Units produced at plant p in month t | Units |
| x[p,m,t] | Units shipped from plant p to U.S. market m in month t | Units |
| y[c,s,p,t] | Components of type c bought from supplier s for plant p, in month t | Units |
| I[p,t] | Inventory at the end of month t at plant p | Units |
| B[m,t] | Stockouts (unmet demand) at market m in month t | Units |

Indices: p – plant (India, China) m – U.S. market (East, West) t – month (1–12)
c – component (Battery, Motor, PCB, Chassis) s – supplier (Local, Regional)

### b) Parameters and Symbols

| Symbol | Description |
|---|---|
| $Cprod_p$ | Unit production cost at plant $p$ (Assembly) |
| $Cship_{p,m}$ | Shipping cost per unit from plant $p$ to market $m$. |
| $Comp_{c,s,p}$ | Cost of purchasing component $c$ from supplier $s$ for plant $p$. |
| $h_p$ | Inventory holding cost per unit at plant $p$. |
| $S_m$ | Stockout (shortage) penalty per unit in market $m$. |
| $Tcomp_{c,s,p}$ | Tariff rate on imported component $c$ from supplier $s$ for a plant $p$. |
| $Tfg_p$ | Tariff rate on finished goods exported from plant $p$. |
| $Eprod_p$ | CO2 emission per production |
| $Eship_{p,m}$ | CO2 emission per shipment from plant $p$ to market $m$. |

| | |
|---|---|
| $CO$ | Carbon price per ton of $CO_2$ emissions. |
| $D_{m,t}$ | Demand in market $m$ during month $t$. |
| $Cap_p$ | Maximum production capacity of plant $p$. |
| $a_c$ | Number of components $c$ required per product. |
| $Supply_{c,s,p,t}$ | Maximum quantity of component $c$ available from supplier $s$ for plant $p$ in month $t$. |
| $E_{max}$ | Maximum allowable total emissions during the planning horizon. |
| $MakeCost_p$ | MakeCost is a parameter and is defined as $Comp_{c,s,p} + Cprod_p$ |

### c) Objective Function

$Z = \sum_t \; [\sum_p \; (Cprod_p * q_{p,t})$ *- assembly cost*

$+ \sum_{p,m} \; (Cship_{p,m} * x_{p,m,t})$ *- shipping cost*

$+ \sum_{p,c,s} \; (Comp_{c,s,p} * y_{c,s,p,t})$ *- component cost*

$+ \sum_p \; (h_p * I_{p,t})$ *- inventory holding cost*

$+ \sum_m (S_m * B_{m,t})$ *- stockout cost* $+\sum_{p,m}$

$(Tfg_p * MakeCost_p * x_{p,m,t})$ *- tariff to import to US*

$+ \sum_{p,c,s} \; (Tcomp_{c,s,p} * Comp_{c,s,p} * y_{c,s,p,t})]$ *- tariff to import components to plant p*

$+ CO(\sum_{p,m,t} Eship_{p,m} x_{p,m,t} + \sum_{p,t} Eprod_p x_{p,m,t})$ *-cost of emission*

### d) Constraints

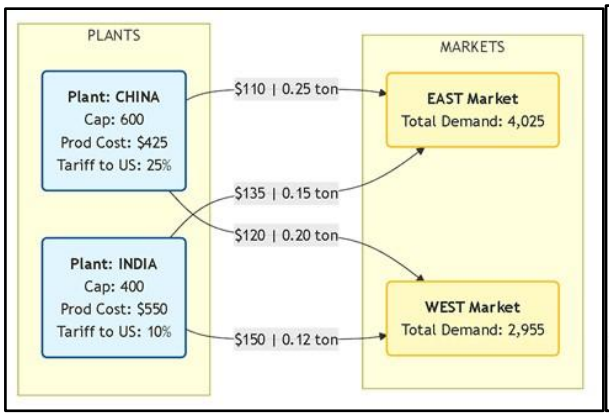
| **Constraint** | **Equation / Explanation** |
|---|---|
| Demand satisfaction | $\sum_p x_{p,m,t} + B_{m,t} = D_{m,t} + B_{m,t-1}$;p={ind,chn},m={east,west}, t={1,2,....,12} |
| Inventory balance | $I_{p,1} = q_{p,1} - \sum_m x_{p,m,1}$ *Initial Inventory Balance*<br>$I_{p,t} = I_{p,t-1} + q_{p,t} - \sum_m x_{p,m,t}$;p={ind,chn},t={1,2,....,12}, m=(east,west} |

| | |
|---|---|
| Plant capacity | $q_{p,t} \leq Cap_p$; p={ind,chn}, t={1,2,....,12} |
| Component requirement | $\sum_s y_{c,s,p,t} \geq a_c * q_{p,t}$; c={bat, mot, pcb, chs}, s={S1, S2}, p={ind,chn}, t={1,2,....,12} |
| Supplier availability | $y_{c,s,p,t} <= Supply_{c,s,p,t}$ c={bat, mot, pcb, chs}, s={S1 , S2}, p={ind,chn}, t={1,2,....,12} |
| Sustainability ($CO_2$ limit) | $\sum_t \sum_{p,m} (Eship_{p,m} * Eprod_p * q_{p,t}) \leq$ Emax; p={ind,chn}, t={1,2,....,12}, m={east,west} |
| Non-negativity | All variables ≥ 0 |

### e) Numerical study

1. **Base Case:** Optimized Production and Distribution Plan

The linear programming model is solved over a 12-month horizon with monthly demand for EAST and WEST markets, plant capacity limits, supplier availability, and a global $CO_2$ cap.




Capacity (Units/Mo): $Cap_p = [600_{CHN} \quad 400_{IND}]$

Unit Production Cost ($): $Cprod_p = [425_{CHN} \quad 550_{IND}]$

Shipping Costs ($): $Cship_{p,m} = \begin{bmatrix} 110 & 120 \\ 135 & 150 \end{bmatrix} \begin{matrix} (CHN) \\ (IND) \end{matrix}$

Emission Factors (Tons): $Eship_{p,m} = \begin{bmatrix} 0.25 & 0.20 \\ 0.15 & 0.12 \end{bmatrix} \begin{matrix} (CHN) \\ (IND) \end{matrix}$

***Fig**. represents the numerical data that is used in the model*

Emax = 4000 (Annual limit in units of CO2emission) CO = 20
(Cost incurred for each unit of CO2 emission) pen(m) = $5000
(Shortage penalty by each unit for each market)

| Comp | BOM | Supplier | Cost (CHN) | Cost(IND) | Tariff (CHN) | Tariff (IND) | Supply Limit |
|---|---|---|---|---|---|---|---|
| **BAT** | 1 | S1 / S2 | $80 / $90 | $70 / $65 | 12% / 15% | 5% / 3% | 1,000 / 800 |
| **MOT** | 1 | S1 / S2 | $60 / $55 | $55 / $58 | 10% / 7.5% | 4% / 6% | 1,200 / 1,000 |

| | | | | | | | |
|---|---|---|---|---|---|---|---|
| **PCB** | 2 | S1 / S2 | $30 / $29 | $28 / $32 | 8% / 10% | 3% / 5% | 1,500 / 1,400 |
| **CHS** | 1 | S1 / S2 | $40 / $45 | $38 / $36 | 9% / 12% | 4% / 6% | 1,500 / 1,400 |

Key Performance Results:

| Metric | Value | Notes / Interpretation |
|---|---|---|
| **Total Cost (Z)** | $7,074,554.50 | Total annual operating cost from objective function |
| **Total Production – IND** | 4,800 units | India is the primary cost-efficient source |
| **Total Production – CHN** | 2,280 units | China acts as flexible capacity for WEST |
| **Total Shipments – EAST** | 4,000 units | Entire EAST demand covered by IND |
| **Total Shipments – WEST** | 3,080 units | Served by both IND and CHN |
| **Total Shortages (B)** | 0 units | 100% demand fulfillment |
| **Emissions** | ≈ 1,415 tons | Below sustainability cap (Emax) |
| **Service Level** | 100% | No backorders or delays |

- India runs at a steady **400** units/month and is the primary low-cost source, serving all **EAST** demand.
- China adjusts output between **100–250** units/month to flexibly meet **WEST** demand.
- A make-to-order system is optimal: no inventory, no holding cost.
- Emissions remain within the sustainability threshold; carbon pricing influences routing but does not constrain feasibility.

2. **Component Tariff Impact:** Effects of Rising Duty Rates on Batteries and PCBs

**Overall System Performance (Battery Tariff + 10%) -**

| Metric | Value | Interpretation |
|---|---|---|
| **Total Cost (Z)** | $7,123,194.50 | ↑ about **$48,640** vs Base Case |

| | | |
|---|---|---|
| **Cost Change vs Base** | +0.7% | Moderate increase driven purely by higher BAT tariffs |
| **Total Production –** | 4,800 units | Same as Base Case (400/month) |
| **IND** | | |
| **Total Production – CHN** | 2,280 units | Same as Base Case |
| **Shipments – EAST** | 4,000 units | All from IND, same pattern as Base Case |
| **Shipments – WEST** | 3,080 units | CHN + IND, same pattern as Base Case |
| **Shortages** | 0 units | Full demand coverage maintained |
| **Inventory** | 0 units | Make-to-order remains optimal |
| **Emissions** | ≈1,415 tons | Unchanged (same flows as Base Case) |

- The model does not change its production, routing, or supplier choices despite the **10%** absolute increase in battery tariffs.
- Because tariffs are raised uniformly across all battery sources and plants, the relative cost advantage between India and China, and between suppliers, remains the same.
- As a result, the optimizer keeps the Base Case structure and simply absorbs a small cost increase (~**0.7%**).
- Service level and sustainability performance remain identical to the Base Case.

**Overall System Performance (PCB Tariff + 20%)-**

| Metric | Value | Interpretation |
|---|---|---|
| **Total Cost (Z)** | $7,153,602.50 | ↑ about **$79,048** vs Base Case |
| **Cost Change vs Base** | +1.1% | Larger impact than battery case (PCB more cost-sensitive here) |
| **Total Production – IND** | 4,800 units | Same as Base Case |

| | | |
|---|---|---|
| **Total Production - CHN** | 2280 units | Same as Base Case |
| **Shipments – EAST** | 4,000 units | Fully served by IND, unchanged |
| **Shipments – WEST** | 3,080 units | CHN + IND, unchanged |
| **Shortages** | 0 units | 100% service level maintained |
| **Inventory** | 0 units | Exact-flow continues |
| **Emissions** | ≈1,415 tons | Unchanged (identical flows) |

- Even with a **20%** absolute increase in PCB tariffs, the model maintains exactly the same production, shipment, and sourcing pattern.
- Again, the tariff is raised uniformly across all PCB options, so the LP has no incentive to switch sourcing or rebalance plants.
- The system remains operationally identical to the Base Case; only the economic outcome changes:
    - Total cost increases by ~**1.1%**.
    - This suggests that the system is robust to uniform component tariff hikes from a structural standpoint, but total cost is sensitive to PCBs somewhat more than to batteries.

**3. Tariff Grid Analysis:** Trade-offs between India and China Export Duties

➢ This scenario varies finished-goods tariff rates for both plants and evaluates total system cost. The heat map shows how changes in tariff policies shift the optimal production balance.

| China/India Tariff | 0.05 | 0.1 | 0.15 | 0.2 |
|---|---|---|---|---|
| 0.1 | $6,305,137.50 | $6,305,137.50 | $6,305,137.50 | $6,305,137.50 |
| 0.15 | $6,603,052.45 | $6,603,183.50 | $6,603,183.50 | $6,603,183.50 |
| 0.2 | $6,836,836.50 | $6,901,186.80 | $6,901,229.50 | $6,901,229.50 |
| 0.25 | $6,929,922.50 | $7,153,602.50 | $7,199,275.50 | $7,199,275.50 |
| 0.3 | $7,023,008.50 | $7,246,688.50 | $7,470,368.50 | $7,497,321.50 |

***Fig**. represents the decision making heat map on the assumption of fluctuating tariff rates*

The heat map shows that total cost rises as either **China's tariff (lower axis)** or **India's tariff (upper axis)** increases. When China's tariff is low (0.10), India's tariff changes have almost no effect because India already operates at full capacity and the sourcing pattern does not change. Once China's tariff exceeds 0.20, India becomes the relatively cheaper source, so cost increases slow down as more supply shifts toward India. The lowest costs occur when both tariffs are low, while the highest costs appear when both countries face high tariffs (≥0.20). Overall, the heat map illustrates how tariff levels shift the sourcing balance between China and India.

**4. Supplier Disruption:** Reduce India's battery supply by 40% during months 5–7 and observe production reallocation and stockouts.

**4.1 Key Metrics – Summary**

| Metric | Value |
|---|---|
| **Total Cost (Z)** | $8,439,965.50 |
| **China Production** | 600 → 180 units (Months 5–7) |
| **India Production** | 125 units/month (full capacity) |
| **Inventory usage** | Inventory built in Months 1–4, exhausted by Month 7 |
| **Total Unmet Demand** | 255 units (East only) |
| **Recovery** | System stabilizes starting Month 8 |

**4.2 Monthly Production – China + India**

- Jan–Apr: China at 600 units/month; India steady at 125.
- May–Jul: China drops to 180 units due to component shortages.
- Aug–Dec: China rebounds to 600 units in August, then adjusts (595 → 475 → 445 → 425) as PCB constraints tighten.

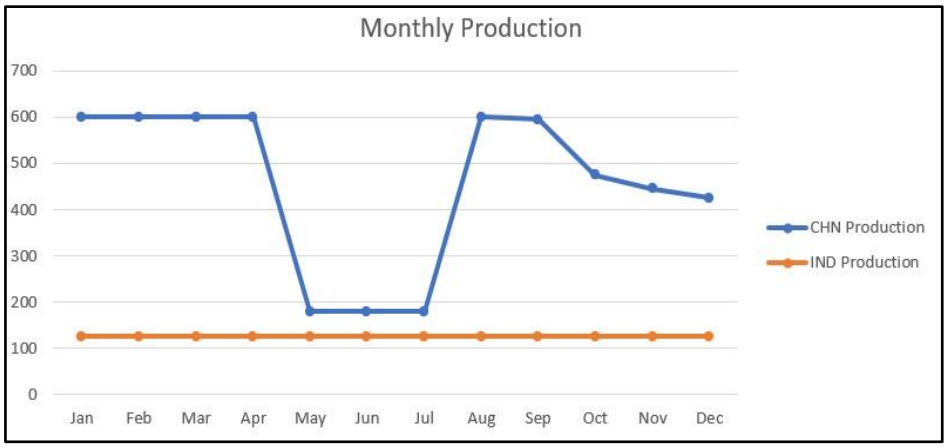



***Fig**. represents the trends in monthly production of each plant*

### 4.3 Shipment Performance – East & West

The WEST market experiences no shortages throughout the disruption period. The EAST market encounters shortages only once early inventory is fully exhausted, showing that initial inventory effectively protects demand for the first six months.

### 4.4 Unmet Demand

Shortages occur only in July and August, after the inventory buffer is depleted; all other months are fully served. The disruption affects the system only briefly, with small and temporary shortages confined to the EAST market. Early inventory delays the impact until Month 7, and normal supply chain operation resumes by Month 9.

**5. Service Level:** Enforce 95% vs. 99% demand fulfillment to evaluate the cost and inventory premium for higher reliability.

| **Metric** | **99% Service Level (High Reliability)** | **95% Service Level (Baseline)** |
|---|---|---|
| **Total Cost (Z)** | $7,003,036 | $6,716,960 |
| **Savings vs Base Case ($7.07M)** | $71,519 | $357,595 |

| | | |
|---|---|---|
| **Total Production – IND** | 4,800 units | 4,800 units |
| **Total Production - CHN** | 2,110 units | 1,831 units |
| **Shipments – EAST** | 4,025 units (from IND) | 4,025 units (from IND) |
| **Shipments – WEST** | 2,885 units | 2,606 units |
| **Inventory** | 0 units | 0 units |

Raising the service level from 95% to 99% greatly reduces shortages (349 → 70 units) but increases total cost and emissions. China's production and WEST shipments rise to support the higher reliability requirement, while India stays at full capacity in both cases. Overall, achieving 99% service requires more production and higher operating cost, illustrating a clear trade-off between reliability, cost, and environmental impact.

**Conclusion:**

NovaRide's dual-plant supply system proves cost-efficient, sustainable, and structurally resilient under baseline and tariff-increase scenarios, with India serving as the dependable low-cost hub and China providing flexible capacity, particularly for the WEST market. While tariff hikes raise total cost modestly, they do not alter the optimal production–shipment structure, indicating strong model stability. However, disruption analysis reveals a strategic vulnerability tied to component shortages at the China plant, leading to short-term unmet demand despite inventory buffering. These findings emphasize the value of supplier diversification, capacity expansion in India, and proactive inventory planning to strengthen long-term resilience and ensure uninterrupted service levels across markets.

**Reference:**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2. Sillekens, T., Koberstein, A., & Suhl, L. (2011). Aggregate production planning in the automotive industry with special consideration of workforce flexibility. *International Journal of Production Research*, *49*(17), 5055-5078.
3. Feylizadeh, M. R., Modarres, M., & Bagherpour, M. (2008). Optimal crashing of multi period–multi product production planning problems. *World Applied Sciences Journal*, 4(4), 499–505.
4. Jamalnia, A., & Soukhakian, M. A. (2009). A hybrid fuzzy goal programming approach with different goal priorities to aggregate production planning. *Computers & Industrial Engineering*, 56(4), 1474–1486.
5. Kazemi Zanjani, M., Nourelfath, M., & Ait-Kadi, D. (2010). A multi-stage stochastic programming approach for production planning with uncertainty in the quality of raw materials and demand. *International Journal of Production Research*, 48(16), 4701–4723.

**Appendices:** Sets p "Plants" / CHN, IND /

m "Markets" / EAST, WEST / c

"Components" / BAT, MOT, PCB, CHS /

s "Suppliers" / S1, S2 / t

"Time period" / 1*12 /;

Table D(m,t) "Demand of market m in time t"

1 2 3 4 5 6 7 8 9 10 11 12

EAST 300 320 310 330 340 350 360 370 355 340 330 320

WEST 200 220 240 230 250 270 260 280 275 260 240 230 ;

Table Cship(p,m) "Cost of shipment from plant p to market m"

EAST WEST

```
CHN 110 120
IND 135 150 ;
Parameter
Cap(p) "Production capacity per plant p in a month"
Ccomp(c,s,p) "Component c purchasing cost from supplier s to plant p"
Cprod(p) "Cost of production for plant p" a(c) "Components needed
per finished product" h(p) "Inventory holding cost per unit" pen(m)
"Shortage penalty cost per unit"
Supply(c,s,p,t) "Max monthly supply of component c"
tau_comp(c,s,p) "Component tariff rate (supplier→plant)"
tau_fg(p) "Finished good tariff rate (plant→US)"
Eship(p,m) "CO2 emission per shipment"
Eprod(p) "CO2 emission per production"
MakeCost(p) "Base making cost for tariff calculation";
Cap("CHN") = 600;
Cap("IND") = 400;
Cprod("CHN") = 425;
Cprod("IND") = 550;
Ccomp("BAT","S1","CHN") = 80; Ccomp("BAT","S2","CHN") = 90;
Ccomp("BAT","S1","IND") = 70; Ccomp("BAT","S2","IND") = 65;
Ccomp("MOT","S1","CHN") = 60; Ccomp("MOT","S2","CHN") = 55;
Ccomp("MOT","S1","IND") = 55; Ccomp("MOT","S2","IND") = 58;
Ccomp("PCB","S1","CHN") = 30; Ccomp("PCB","S2","CHN") = 29;
Ccomp("PCB","S1","IND") = 28; Ccomp("PCB","S2","IND") = 32;
Ccomp("CHS","S1","CHN") = 40; Ccomp("CHS","S2","CHN") = 45;
Ccomp("CHS","S1","IND") = 38; Ccomp("CHS","S2","IND") = 36;
a("BAT") = 1; a("MOT") = 1; a("PCB") = 2; a("CHS") = 1;
```

```
MakeCost(p) = Cprod(p) + SUM((c,s), Ccomp(c,s,p));

h("CHN") = 5; h("IND") = 5; pen("EAST") = 5000;

pen("WEST") = 5000;

Loop(t,

Supply("BAT","S1","CHN",t) = 300;

Supply("BAT","S2","CHN",t) = 400;

Supply("BAT","S1","IND",t) = 150;

Supply("BAT","S2","IND",t) = 100;

Supply("MOT","S1","CHN",t) = 500;

Supply("MOT","S2","CHN",t) = 400;

Supply("MOT","S1","IND",t) = 400;

Supply("MOT","S2","IND",t) = 350;

Supply("PCB","S1","CHN",t) = 600;

Supply("PCB","S2","CHN",t) = 600;

Supply("PCB","S1","IND",t) = 150;

Supply("PCB","S2","IND",t) = 100;

Supply("CHS","S1","CHN",t) = 600;

Supply("CHS","S2","CHN",t) = 600;

Supply("CHS","S1","IND",t) = 200; Supply("CHS","S2","IND",t)

= 300;

);

*-----------------------------------------------*

* Supplier Disruption Scenario:

* India loses 40% of BAT supply in Months may, june & july

*-----------------------------------------------*

$ontext

Loop(t$(ord(t)>=5 and ord(t)<=7),
```

```
Supply("BAT","S1","CHN",t) = 0.6 * Supply("BAT","S1","CHN",t);
Supply("BAT","S2","CHN",t) = 0.6 * Supply("BAT","S2","CHN",t);
Supply("PCB","S1","CHN",t) = 0.3 * Supply("PCB","S1","CHN",t);
Supply("PCB","S2","CHN",t) = 0.3 * Supply("PCB","S2","CHN",t);
);
$offtext tau_comp("BAT","S1","CHN")
= 0.12; tau_comp("BAT","S2","CHN")
= 0.15; tau_comp("BAT","S1","IND")
= 0.05; tau_comp("BAT","S2","IND")
= 0.03; tau_comp("MOT","S1","CHN")
= 0.10; tau_comp("MOT","S2","CHN")
= 0.075;
tau_comp("MOT","S1","IND") = 0.04;
tau_comp("MOT","S2","IND") = 0.06;
tau_comp("PCB","S1","CHN") = 0.08;
tau_comp("PCB","S2","CHN") = 0.10;
tau_comp("PCB","S1","IND") = 0.03;
tau_comp("PCB","S2","IND") = 0.05;
tau_comp("CHS","S1","CHN") = 0.09;
tau_comp("CHS","S2","CHN") = 0.12;
tau_comp("CHS","S1","IND") = 0.04;
tau_comp("CHS","S2","IND") = 0.06;
tau_fg("CHN") = 0.25; tau_fg("IND") =
0.10;
*-------------------------------------------------------------*
* EMISSIONS
*-------------------------------------------------------------*
```

```
Eship("CHN","EAST") = 0.25; Eship("CHN","WEST") = 0.20;

Eship("IND","EAST") = 0.15; Eship("IND","WEST") = 0.12;

Eprod("CHN") = 0.30;

Eprod("IND") = 0.25;

Scalar

Emax "Annual emission limit" /4000/

CO "Carbon price per ton of CO2" /20/;
*=====================================================================*

* DECISION VARIABLES

*=====================================================================*

Variables q(p,t) "Units produced"

x(p,m,t) "Units shipped" y(c,s,p,t)

"Components purchased"

I(p,t) "Inventory"

B(m,t) "Unmet demand"

z "Total cost";

Positive Variables q, x, y, I, B;

Equations

DemandSatisfaction(m,t) "Demand constraints"

InitialBacklog(m) "Initial backlog"

ClearBacklog(m) "clear backlogs"

InventoryBalance1(p) "Opening inventory"

InventoryBalance(p,t) "Inventory flow"

PlantCap(p,t) "Production capacity"

SupplierAvail(c,s,p,t) "Supplier limits"

ComponentRequirement(c,p,t) "BOM requirements"
```

```
Sustainability "Emission cap" Objective

"Total cost minimization";

Objective.. z

=e= SUM(t,

SUM(p, Cprod(p) * q(p,t))

+ SUM((p,m), Cship(p,m) * x(p,m,t))

+ SUM((p,c,s), Ccomp(c,s,p) * y(c,s,p,t))

+ SUM(p, h(p) * I(p,t))

+ SUM(m, pen(m) * B(m,t))

+ SUM((p,m), tau_fg(p) * MakeCost(p) * x(p,m,t))

+ SUM((c,s,p), tau_comp(c,s,p) * Ccomp(c,s,p) * y(c,s,p,t))

)

+ CO * ( SUM((p,m,t), Eship(p,m)*x(p,m,t))

+ SUM((p,t), Eprod(p) *q(p,t))

);

* Rolling Backlog Demand Constraint

DemandSatisfaction(m,t)..

SUM(p, x(p,m,t)) + B(m,t) =e= D(m,t) + (B(m,t-1)$(ord(t)>1));

InitialBacklog(m)..

B(m,'1') =e= 0;

ClearBacklog(m)..

B(m,'12') =e= 0;

InventoryBalance1(p)..

I(p,'1') =e= q(p,'1') - SUM(m, x(p,m,'1')); InventoryBalance(p,t)$(ord(t)

> 1)..

I(p,t) =e= I(p,t-1) + q(p,t) - SUM(m, x(p,m,t));
```

```
PlantCap(p,t).. q(p,t) =l= Cap(p);

ComponentRequirement(c,p,t).. SUM(s,

y(c,s,p,t)) =e= a(c) * q(p,t);

SupplierAvail(c,s,p,t)..

y(c,s,p,t) =l= Supply(c,s,p,t);

Sustainability..

SUM((p,m,t), Eship(p,m) * x(p,m,t))

+ SUM((p,t), Eprod(p) * q(p,t))

=l= Emax;

Model Project /all/;

Solve Project using lp minimizing z;

Display z.l, q.l, x.l, I.l, B.l, y.l;
```